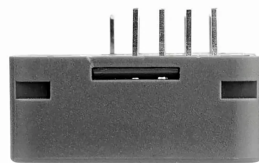
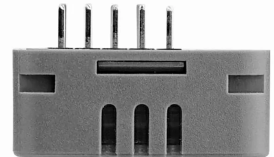
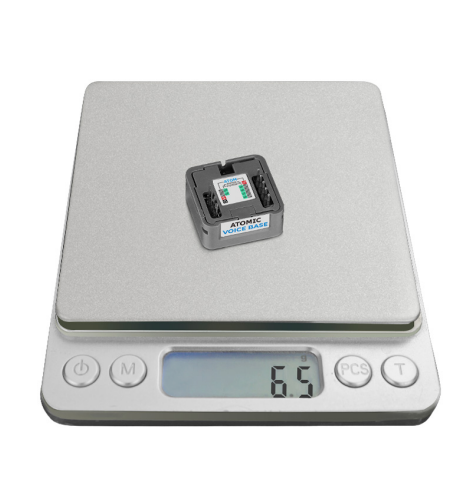


Atomic Voice Base

SKU:A149

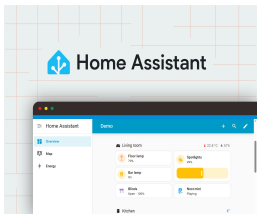




Description

Atomic Voice Base is a **voice recognition** base designed specifically for the M5 Atom series hosts. It integrates the ES8311 mono audio decoder, MEMS microphone, and NS4150B power amplifier. The ES8311 provides 24-bit audio resolution and a sampling rate of 16KHz to 64KHz, supporting I2S communication and high-fidelity audio processing. The MEMS **microphone** with high signal-to-noise ratio and digital output is suitable for automatic voice recognition, while the built-in **speaker** ensures efficient audio output. The device supports **full-duplex** communication, allowing simultaneous sound transmission and reception, enhancing interactive functions such as voice recognition, wake-up, and recording playback, making it ideal for applications in smart homes and education.

Tutorial



Home Assistant

This tutorial introduces how to combine AtomS3R + Atomic Voice Base to integrate voice assistant functions into Home Assistant.

Features

- Voice recognition
- Mono audio codec
- MEMS microphone

- Class D amplifier
- Speaker
- Development platforms
 - UiFlow1
 - UiFlow2
 - Arduino IDE

Includes

- 1 x Atomic Voice Base

Applications

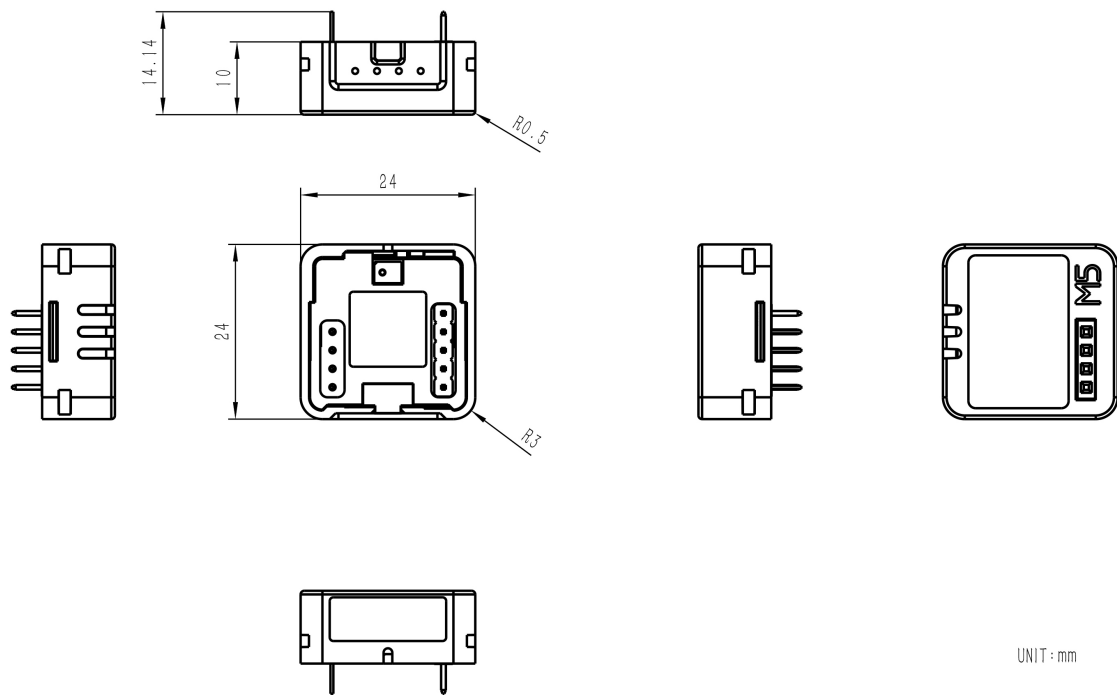
- AI voice recognition interaction systems
- Smart home control

Specifications

Specification	Parameters
Audio Codec	ES8311: 24-bit resolution, I2S protocol
MEMS Microphone	MSM381A3729H9BPC, SNR: ≥65 dB
Power Amplifier	NS4150B: Class D power amplifier
Speaker	2014 cavity speaker: 1W@8Ω
Standby Current	DC3.3V/3.36mA
Operating Current	DC3.3V/14.84mA
Operating Temperature	0 ~ 40°C
Product Size	24.0 x 24.0 x 14.1mm
Product Weight	6.5g
Package Size	85.0 x 65.5 x 15.0mm
Gross Weight	12.5g

Schematics

- [Atomic Voice Base Schematic PDF](#)



Datasheets

- [ES8311](#)
- [MEMS MIC](#)
- [NS4150B](#)

Softwares

MIC Sampling Rate

The Atomic Voice Base microphone supports a sampling rate range of 16KHz-64KHz

Quick Start

- [OpenAI Voice Assistant For AtomS3R + Atomic Voice Base Tutorial](#)
- [Atomic Voice Base XiaoZhi Voice Assistant](#)
- [AtomS3R + Atomic Voice Base Voice Assistant](#)

Arduino

- [Atomic Voice Base Arduino Library](#)

UiFlow1

coming soon

UiFlow2

- [Atomic Voice Base UiFlow2 Docs](#)

ESP-IDF

- [Open RealtimeAPI Embedded SDK](#)

Video

- Atomic Voice Base Product Introduction and Case Demonstration

[Atomic Echo Base.mp4](#)

- OpenAI Voice Assistant For AtomS3R + Atomic Voice Base

[OpenAI Voice Assistant For AtomS3R.mp4](#)

Product Comparison

Product Comparison Table		
		
	Atomic Voice Base	Atom Voice
Solution	Base (No SoC)	Built-in SoC (ESP32-PICO-D4)
MIC	MSM381A3729H9BPC	SPM1423
Amplifier	NS4150B@Class D amplifier	NS4150B@Class D amplifier
Speaker	1W@8Ω	0.5W@8Ω
Audio Sampling Rate	16KHz-64KHz	8kHz-96kHz
Programmable RGB/Button	/	SK6812
Communication Mode	Full-duplex	Half-duplex